Biochemistry Cume

Total points: 100 points

1. Give reason(s) as to why the V_{max} and K_M values of an enzyme, if calculated directly from the Michaelis – Menten curve, are not accurate. (**10 points**)

2. What is a Suicide inhibitor. Give an example. (**15 points**)

3. The enzyme urease enhances the rate of urea hydrolysis at pH 8.0 and 20 °C by a factor of 10^{14}. If a given quantity of urease can completely hydrolyze a given quantity of urea in 5 minutes at 20 °C and pH 8.0, how long would it take for this amount of urea to be hydrolyzed under the same conditions in the absence of urease. (**20 minutes**).

4. Carbonic anhydrase (molecular mass =30 kDa), present in erythrocytes, has one of the highest turnover numbers among known enzymes. It catalyzes the reversible conversion hydration of CO_2:

$$H_2O + CO_2 \leftrightarrow H_2CO_3$$

This is an important process in the transport of CO_2 from the tissues to the lungs. If 10 μg of pure carbonic anhydrase catalyzes the hydration of 0.30 g of CO_2 in 1 minute at 37 °C at V_{max}, what is the turnover number (k_{cat}) of carbonic anhydrase (in min^{-1})? (**25 points**)

5. **A**. At what substrate concentration will an enzyme with a V_{max} of 40 sec^{-1} and a K_M of 0.005 M show one-quarter of its maximum rate? (**15 points**)

B. Determine the fraction of V_{max} that would occur when the substrate concentration [S] is twenty times the K_M of the enzyme. (**15 points**)